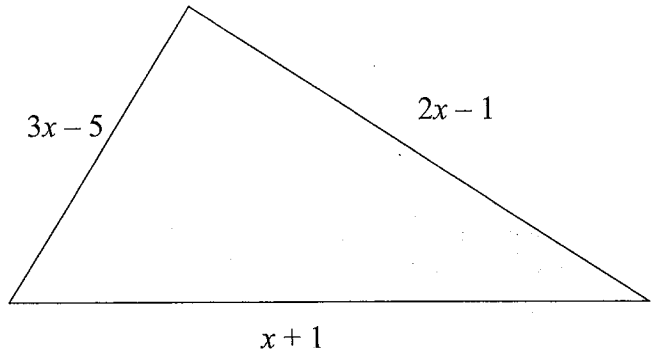
1 The lengths, in cm, of the sides of a triangle are $3x - 5$, $2x - 1$ and $x + 1$



(a) Write down an expression, in terms of x, for the perimeter of the triangle.

$3x - 5 + 2x - 1 + x + 1$

$6x - 5$ cm

(2)

The perimeter of the triangle is 31 cm.

(b) Work out the value of x.

$6x - 5 = 31$

$6x = 36$

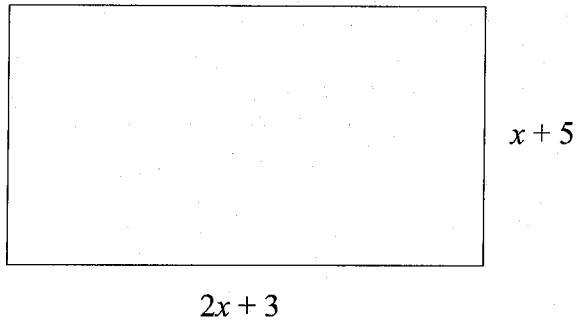
$x = 6$

6 [cm]

(2)

(Total for question 1 is 4 marks)

2 A rectangle has a length of $(2x + 3)$ cm and a width of $(x + 5)$ cm.



(a) Find an expression for the perimeter of the rectangle.

$2(2x+3) + 2(x+5)$

$4x + 6 + 2x + 10$

$6x + 16$

$6x + 16$ cm

(2)

(b) Given the rectangle has a perimeter of 43 cm find the value of x.

$6x + 16 = 43$

$6x = 27$

$x = \frac{27}{6} = \frac{9}{2} = 4.5$

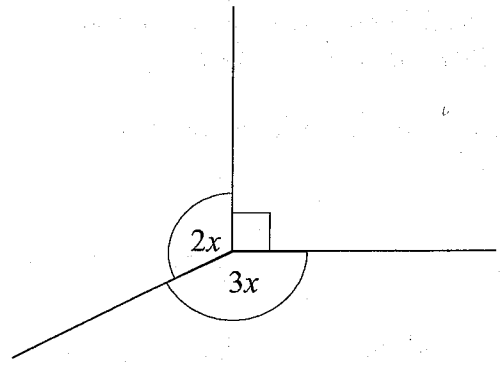
4.5 [cm]

(2)

(Total for question 2 is 4 marks)

(or $\frac{9}{2}$)

3



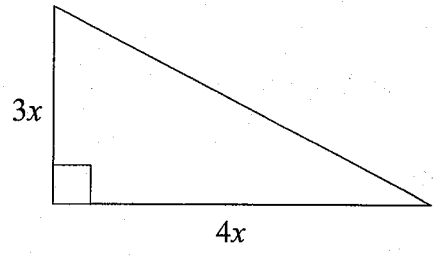
Find the value of x.

$$2x + 3x + 90 = 360$$
$$5x + 90 = 360$$
$$5x = 270$$
$$x = 54$$

54°

(Total for question 3 is 3 marks)

4 The diagram shows a right angled triangle.



The area of the triangle is 294 cm^2

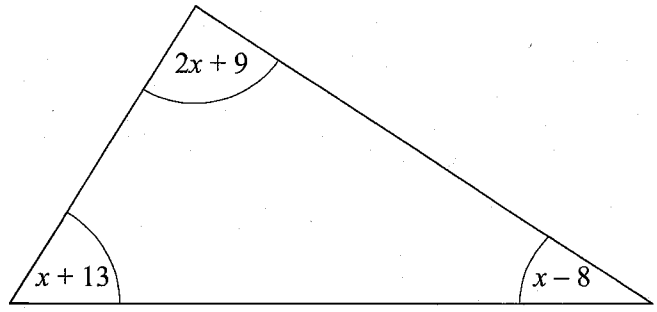
Work out the value of x.

$$\frac{1}{2} \times 4x \times 3x = 294$$
$$\frac{1}{2} \times 12x^2 = 294$$
$$6x^2 = 294$$
$$x^2 = \frac{294}{6} = \frac{147}{3} = 49$$
$$x = \sqrt{49}$$
$$= 7$$

7 [cm]

(Total for question 4 is 3 marks)

5 The sizes of the angles, in degrees, of a triangle are $2x + 9$, $x + 13$ and $x - 8$



Work out the value of x.

$$2x + 9 + x - 8 + x + 13 = 180$$
$$4x + 14 = 180$$
$$4x = 166$$
$$x = \frac{166}{4} = \frac{83}{2} = 41.5$$

41.5 OR $\frac{83}{2}$

(Total for question 5 is 3 marks)

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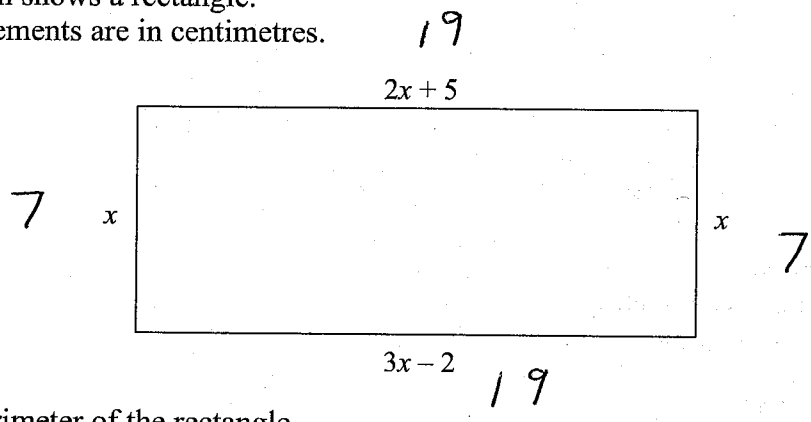
6 The diagram shows a rectangle.
All measurements are in centimetres.



Find the perimeter of the rectangle.

$$2x + 5 = 3x - 2$$
$$5 = x - 2$$
$$7 = x$$

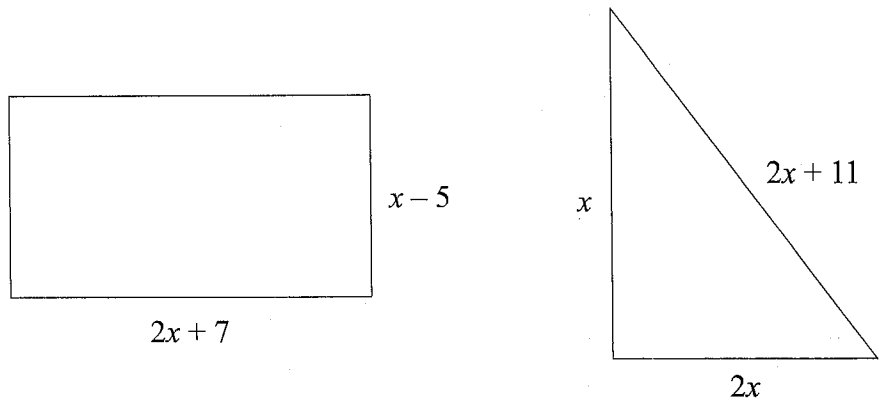
$$2(7) + 5 = 19$$

$$2(19) + 2(7)$$
$$38 + 14$$

52 cm

(Total for question 6 is 3 marks)

7 The diagram shows a rectangle and a triangle.



The perimeter of the rectangle is equal to the perimeter of the triangle.

Find the value of x.

$2(x-5) + 2(2x+7) = x + 2x + 2x + 11$

$2x - 10 + 4x + 14 = 5x + 11$

$6x + 4 = 5x + 11$

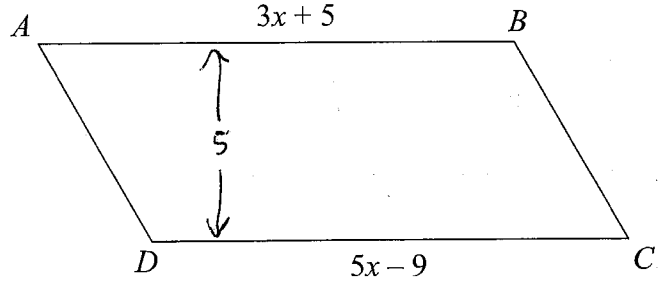
$x + 4 = 11$

$x = 7$

7

(Total for question 7 is 3 marks)

8



ABCD is a parallelogram
All measurements are in centimetres.
The perpendicular height of the parallelogram is 5 cm.

Find the area of *ABCD*

$3x + 5 = 5x - 9$

$5 = 2x - 9$

$14 = 2x$

$x = 7$

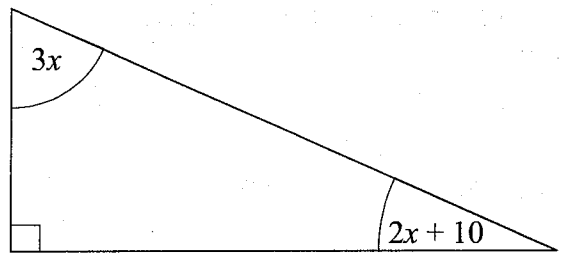
$3(7) + 5 = 26$

26×5

130 cm^2

(Total for question 8 is 4 marks)

9 The diagram shows a right-angled triangle.
All of the angles are in degrees.



Find the value of size of the smallest angle in the triangle.

$3x + 2x + 10 + 90 = 180$

$5x + 100 = 180$

$5x = 80$

$x = 16$

$3(16) = 48 \qquad 2(16) + 10 = 42$

42 °

(Total for question 9 is 3 marks)

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10 Adam has some marbles.
Bradley has twice as many marbles are Adam.
Chris has 5 more marbles than Bradley.

In total they have 55 marbles.

How many marbles does Chris have?

Adam $= x$

Bradley $= 2x$

Chris $= 2x + 5$

$x + 2x + 2x + 5 = 55$

$5x + 5 = 55$

$5x = 50$

$x = 10$

$2(10) + 5 = 25$

25

(Total for question 10 is 3 marks)

$3x$ x

11 The size of the largest angle in a triangle is three times the size of the smallest angle.
The other angle is 35° more than the smallest angle.

$x + 35$

Work out, in degrees, the size of each angle in the triangle.
You must show your working.

$3x + x + x + 35 = 180$

$5x + 35 = 180$

$5x = 145$

$x = 29$

$29 + 35 = 64$

$3(29) = 87$

29 °, 64 °, 87 °

(Total for question 11 is 5 marks)

$3x$ x

12 Lucy is three times as old as Alex.
Lucy is 7 years older than Megan.
The sum of their ages is 126.

Find the ratio of Alex's age to Lucy's age to Megan's age.

Alex $= x$

Lucy $= 3x$

Megan $= 3x - 7$

$x + 3x + 3x - 7 = 126$

$7x - 7 = 126$

$7x = 133$

$x = 19$

$3(19) = 57$

$57 - 7 = 50$

19 : 57 : 50

(Total for question 12 is 4 marks)